

I. Fundamentals of Cross-Validation

Q1: What is the main purpose of Cross-Validation (CV) as a statistical approach in Machine Learning?

A1: To evaluate and estimate the **generalization performance** of an ML model on unseen data.

Q2: CV provides an estimate of the generalizability of an ML model by considering what type of varied data?

A2: **Different subsets of the available dataset**, typically through multiple training/validation splits.

Q3: What is the Validation Set used for during the CV process?

A3: To **assess model performance** on unseen data and tune hyperparameters.

Q4: Why is validation necessary during model training in relation to model fitting?

A4: To **prevent overfitting** and ensure the model does not memorize the training data.

Q5: What two benefits does CV provide by making efficient use of limited data?

A5: **Maximizes data usage for training** and **provides a robust estimate of model performance**.

Q6: CV is often carried out in conjunction with the tuning of what type of model settings?

A6: **Hyperparameters** (e.g., regularization strength, tree depth, learning rate).

Q7: During CV, how is the actual training data for a given fold defined?

A7: All data **except the chosen validation fold** from the original training set.

II. K-Fold Cross-Validation

Q8: What is the method used in k-fold cross-validation to partition the initial training set?

A8: The dataset is split into **k equal-sized folds**.

Q9: If the training set has (n) datapoints and (k) folds are used, how many datapoints are in each fold?

A9: Approximately (n/k) datapoints per fold.

Q10: In k-fold CV, how many folds are used to train the ML model in any single iteration?

A10: **k-1 folds** are used for training.

Q11: How is the remaining fold used in the k-fold CV process?

A11: It is used as the **validation set** for that iteration.

Q12: After a complete k-fold CV process, how many times would every datapoint have participated in training and validation?

A12: Each datapoint is used **exactly once for validation** and $(k-1)$ times for training.

Q13: What is the common name given to $k=5$ fold cross-validation?

A13: **5-Fold Cross-Validation**.

Q14: How is the average model performance metric estimated from a complete k-fold CV run?

A14: By computing the **mean of the performance metric** across all k folds.

Q15: What type of CV should be used in classification problems with imbalanced classes?

A15: **Stratified k-fold CV**, which preserves class proportions in each fold.

III. Leave-One-Out Cross-Validation (LOOCV)

Q16: In LOOCV, how many folds are used relative to the number of datapoints?

A16: **n folds**, one for each datapoint.

Q17: For a dataset with n datapoints, how many are used to train the model in each LOOCV iteration?

A17: **$n-1$ datapoints**.

Q18: How is the validation set defined for a single iteration of LOOCV?

A18: The **single datapoint left out** of the training set.

Q19: What effect does using a large training data in each LOOCV fold have on bias?

A19: It **reduces bias**, since almost all data is used for training.

Q20: What effect does LOOCV have on the variability of the model performance metrics?

A20: It **increases variance**, since each validation set is only one datapoint.

Q21: Why is LOOCV considered computationally expensive?

A21: Because the model must be **trained n times**, once for each datapoint.

Q22: When is LOOCV preferred over k-fold CV?

A22: When the **dataset is very small**, and maximum training data is needed for each iteration.

IV. Repeated Random Subsampling CV

Q23: What is the third type of cross-validation technique besides k-fold and LOOCV?

A23: **Repeated Random Subsampling CV** (also called Monte Carlo CV).

Q24: In Repeated Random Subsampling CV, how many times can a datapoint be selected for the validation set?

A24: **Multiple times**, as splits are random and repeated.

Q25: How is the training process different in Repeated Random Subsampling CV compared to k-fold CV?

A25: Training and validation sets are **randomly selected in each iteration**, not systematically partitioned.

Q26: What is the key advantage of Repeated Random Subsampling CV over k-fold CV?

A26: It allows a **larger number of possible train/validation splits**, increasing robustness of performance estimates.

V. CV Metrics and Interpretation

Q27: What are three examples of common performance metrics estimated from CV?

A27: **R^2 , RMSE, MAE.**

Q28: When monitoring training and validation loss, what indicates overfitting?

A28: **Low training loss** but **high validation/test loss**.

Q29: When monitoring training and validation loss, what indicates underfitting?

A29: **High training and validation loss**, showing poor fit to data.

Q30: What property of a dataset is leveraged by Stratified k-fold in classification?

A30: **Class label distribution**, to preserve class proportions in each fold.